

Q17. There are 10 emails total: 7 Normal & 3 Spam

$$P(\text{Normal}) = \frac{7}{10} = 0.7$$

$$P(\text{Spam}) = \frac{3}{10} = 0.3$$

Word counts

<u>Word</u>	<u>Normal count</u>	<u>Spam count</u>
Semester	9	0
Student	6	1
Project	3	8
Money	1	4
	19	13

a) Conditional probability of each word for each class:

$$P(\text{Semester} | \text{Normal}) = \frac{9}{19} = 0.47 \quad P(\text{Semester} | \text{Spam}) = \frac{0}{13} = 0$$

$$P(\text{Student} | \text{Normal}) = \frac{6}{19} = 0.32 \quad P(\text{Student} | \text{Spam}) = \frac{1}{13} = 0.08$$

$$P(\text{Project} | \text{Normal}) = \frac{3}{19} = 0.16 \quad P(\text{Project} | \text{Spam}) = \frac{8}{13} = 0.62$$

$$P(\text{Money} | \text{Normal}) = \frac{1}{19} = 0.05 \quad P(\text{Money} | \text{Spam}) = \frac{4}{13} = 0.31$$

$$\text{Here } P(\text{word} | \text{Class}) = \frac{\text{Count of the word in that class}}{\text{Total no. of words in that class}}$$

b) Probabilities: $P(\text{Normal} | \text{Email})$ and $P(\text{Spam} | \text{Email})$

$$P(\text{Normal} | \text{Email}) = P(\text{Normal}) \cdot P(\text{Semester} | \text{Normal}) \cdot P(\text{Project} | \text{Normal}) \cdot (P(\text{Money} | \text{Normal}))^2$$

$$= \left(\frac{7}{10}\right) \left(\frac{9}{19}\right) \left(\frac{3}{19}\right) \left(\frac{1}{19}\right)^2 = 0.000146$$

$$P(\text{Spam} | \text{Email}) = P(\text{Spam}) \cdot P(\text{Semester} | \text{Spam}) \cdot P(\text{Project} | \text{Spam}) \cdot (P(\text{Money} | \text{Spam}))^2$$

$$= \left(\frac{3}{10}\right) \left(\frac{0}{13}\right) \left(\frac{8}{13}\right) \left(\frac{4}{13}\right)^2 = 0$$

b) 2) No, you cannot reliably predict. The probability of Spam class is 0. This is known as zero frequency problem. The word "Semester" never appeared in the spam training data, the model assigns it a 0% chance of being a spam email.

c) 1) We can solve this by using Laplace smoothing (Add-1 smoothing). We add 1 to every word count (the numerator). To balance this, we add k to the total word count (the denominator), where k is the number of unique words in our vocabulary.

Here, $k = 4$

New Probabilities with Add-1 Smoothing:

$$P(\text{Semester} | \text{Normal}) = \frac{(9+1)}{(19+4)} = \frac{10}{23} \quad P(\text{Semester} | \text{Spam}) = \frac{(0+1)}{(13+4)} = \frac{1}{17}$$

$$P(\text{Student} | \text{Normal}) = \frac{(6+1)}{(19+4)} = \frac{7}{23} \quad P(\text{Student} | \text{Spam}) = \frac{(1+1)}{(13+4)} = \frac{2}{17}$$

$$P(\text{Project} | \text{Normal}) = \frac{4}{23} \quad P(\text{Project} | \text{Spam}) = \frac{9}{17}$$

$$P(\text{Money} | \text{Normal}) = \frac{2}{23} \quad P(\text{Money} | \text{Spam}) = \frac{5}{17}$$

$$P(\text{Normal} | \text{Email}) = \left(\frac{7}{10}\right) \left(\frac{10}{23}\right) \left(\frac{4}{23}\right) \left(\frac{2}{23}\right)^2 = 0.0004$$

$$P(\text{Spam} | \text{Email}) = \left(\frac{3}{10}\right) \left(\frac{1}{17}\right) \left(\frac{9}{17}\right) \left(\frac{5}{17}\right)^2 = 0.0008$$

3) By using Laplace smoothing, this email is Spam because $0.0008 > 0.0004$.